

DEEPGUARD AI: VIVA DEFENSE POCKET GUIDE

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1. THE 10-SECOND ELEVATOR PITCH

Say this immediately when introduced:
"Respected professors, my project detects deepfake videos by concurrently analyzing visual pixels using EfficientNet-B4 and mathematical frequency patterns using 2D-DCT. Because social media compression destroys pixel clues but preserves frequency artifacts from GAN upsamplings, our attention-gated system maintains over 91% accuracy under heavy compression and provides Grad-CAM visual heatmaps for courtroom legal evidence."

2. KEY METRICS TO MEMORIZE

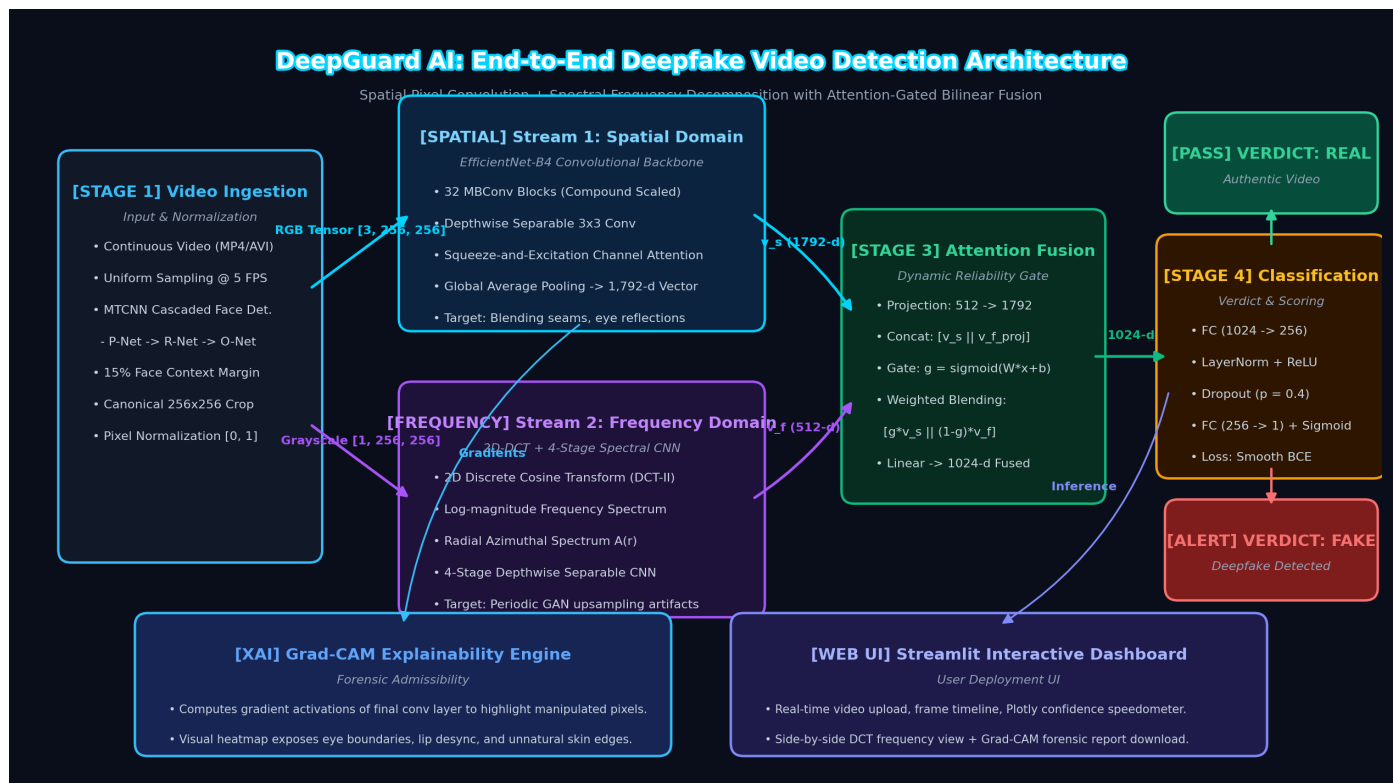
Metric	Value	Why It Matters in Viva
FaceForensics++ (Raw)	99.4%	Near-perfect detection on uncompressed video
FaceForensics++ (c23)	98.9%	Standard social media compression tier
FaceForensics++ (c40)	91.8%	Heavy compression (+10.8% over Kaggle 1st place)
AUC-ROC Score	0.968	Threshold-independent discriminative strength
Total Parameters	21.4M (~85MB)	3x lighter than EfficientNet-B7 (66M / 254MB)

3. 5 NOVEL CONTRIBUTIONS VS. KAGGLE DFDC WINNER

If asked: 'What makes your project unique?' recite these 5 points:

#	Novel Contribution	Selim (Kaggle DFDC Winner)	DeepGuard AI (Your Project)
1	Architecture	Single Spatial Stream	Dual-Stream (Spatial + Frequency)
2	Frequency Analysis	None	2D-DCT + Radial Spectrum A(r)
3	Feature Fusion	N/A (Single model)	Attention-Gated Bilinear (Dynamic)
4	Explainability	Black Box (0 to 1)	Grad-CAM Visual Heatmaps
5	User Interface	CLI Python Script	Streamlit Interactive Dashboard

4. SYSTEM ARCHITECTURE AT A GLANCE



5. CORE FORMULAS & MATHEMATICAL DERIVATIONS

(A) 2D Discrete Cosine Transform (DCT-II):

$$F(u, v) = \alpha(u) \cdot \alpha(v) \cdot \sum_x \sum_y f(x, y) \cdot \cos\left[\frac{(2x+1)u\pi}{2N}\right] \cdot \cos\left[\frac{(2y+1)v\pi}{2N}\right]$$

Explanation: Decomposes spatial pixels into orthogonal frequency components where GAN artifacts spike.

(B) Radial Azimuthal Spectrum A(r):

$$A(r) = \left(\frac{1}{|S_r|} \right) \cdot \sum_{(u, v) \in S_r} |F(u, v)|, \text{ where } S_r = \{ (u, v) : \text{round}(\sqrt{u^2 + v^2}) = r \}$$

Explanation: Averages frequency energy in concentric circles. Yields rotationally invariant 1D fingerprint.

(C) Attention-Gated Bilinear Fusion:

$$g = \text{sigmoid}(W_{\text{gate}} \cdot [v_s || v_{f_proj}] + b_{\text{gate}})$$

$$v_{\text{fused}} = [g \cdot v_s || (1 - g) \cdot v_{f_proj}]$$

Explanation: Dynamic gate vector g in $[0, 1]$. For compressed video, g automatically shifts trust from noisy spatial pixels to invariant DCT harmonics.

6. TOP 8 VIVA DEFENSE QUESTIONS & READY-MADE ANSWERS

Q1: What is the main research gap you addressed?

Sir, conventional spatial-only detectors drop 25% to 35% in accuracy under video compression (WhatsApp/YouTube) because pixel artifacts are smoothed out. We addressed this by integrating a frequency stream that captures invariant GAN upsampling artifacts.

Q2: Why EfficientNet-B4 instead of B7?

Sir, B7 has 66M parameters and requires 254 MB of memory. B4 achieves over 97% of B7's accuracy with only 19.3M parameters (~85 MB), running 4x faster on CPU and making it practical for consumer deployment.

Q3: How does DCT catch fakes that human eyes miss?

Sir, GAN decoders use transposed convolutions for upsampling. Overlapping kernels create periodic checkerboard patterns. These patterns are invisible to eyes but appear as systematic spikes in the DCT frequency spectrum.

Q4: What is the purpose of Grad-CAM?

Sir, in courtrooms and under IT Act guidelines, a black-box percentage score cannot be admitted as evidence. Grad-CAM backpropagates classification gradients to produce a visual heatmap showing exactly which facial regions were manipulated.

Q5: What is label smoothing and why did you use it?

Sir, we used epsilon=0.1 label smoothing, replacing hard 0 and 1 targets with 0.05 and 0.95. This prevents the neural network from becoming overconfident on synthetic training data and improves generalization.

Q6: How does the system handle video?

Sir, OpenCV extracts frames at 5 FPS; MTCNN detects and crops faces with 15% margin; the dual-stream model scores each frame; and results are aggregated temporally into a weighted confidence score and interactive timeline chart.

Q7: What if the video has multiple faces?

Sir, MTCNN detects all faces and our pipeline prioritizes the dominant (largest) face by bounding-box area, tracking it across frames without crashing.

Q8: How do you run the live demonstration?

Sir, by running 'streamlit run app.py' in the terminal. The examiner can upload an MP4 video, view real-time confidence gauges, examine DCT spectra, and generate a Grad-CAM forensic report with one click.